A / B testing

Learning Objectives

By the end of this session, you will be able to:

- Understand A/B testing
- Confidence interval role in hypothesis testing
- Concept of p-values in statistical tests

A/B Testing – the popular use case

Scenario
Your company has launched a new website design (Version B) and wants to know if it leads to more conversions compared to the old design (Version A).

The Goal: Use A/B testing to determine if the new design improves the conversion rate.

Sample 1 (old design A): n1=5013, x1=250 conversions
Sample 2 (new design B): n2=4987, x2=325 conversions

At **5% level of significance**, test whether there is a **significant difference** in conversion rates between the two designs.
Let's solve it?

Two proportion test

Two independent sample – Large Sample(>=30)

Test Statistics

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

- The sample proportions are calculated as:

$$\hat{p}_1 = \frac{x_1}{n_1}$$

$$\hat{p}_2 = \frac{x_2}{n_2}$$

- The **pooled proportion p-hat is** calculated as:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

A/B Testing – calculations?

$$\hat{p}_1 = \frac{x_1}{n_1}$$

$$\hat{p}_2 = \frac{x_2}{n_2}$$

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Sample 1 (old design A): n1=5013, x1=250 conversions

Sample 2 (new design B): n2=4987, x2=325 conversions

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

A/B Testing – conclusion

- **Calculated Z-value = -3.29**
- **Critical Z-value = ±1.96**
- Since **|Zcalculated|=3.29>Zcritical=1.96**, we **reject the null hypothesis H0.**
- There is **a statistically significant difference** in the proportion of conversion rate for the two designs.
- This suggests that **website design has an** influence on conversion rates.

Confidence interval and p-values

What is a Confidence Interval in Hypothesis Testing?

A **confidence interval (CI)** is a range of values that is likely to contain the **true population parameter** (like the mean or proportion) with a certain level of confidence (typically 95%).

Instead of saying, “The average weight is 70 kg,” we say:
👉 “We are 95% confident the average weight lies between 67 and 73 kg.”

It provides more information than a point estimate and helps in decision-making — if your hypothesized value (e.g., H_0 mean) falls **outside** the interval, it’s likely **not true**.

Formula

Z Confidence Interval (when population standard deviation is known)

$$CI = \bar{x} \pm Z\left(\frac{\sigma}{\sqrt{n}}\right)$$

Where:

- $\bar{x}$ = sample mean
- Z = Z-score for desired confidence level (e.g., 1.96 for 95%)
- σ = population standard deviation
- n = sample size

t Confidence Interval (when population standard deviation is unknown)

$$CI = \bar{x} \pm t\left(\frac{s}{\sqrt{n}}\right)$$

Where:

- t = t-value based on degrees of freedom (df = n−1)
- s = sample standard deviation

Solved Example – Z-Test CI

Problem:
A sample of 100 students has an average score of 75. The population standard deviation is 10. Find the **95% confidence interval** for the true mean.

$$CI = \bar{x} \pm Z\left(\frac{\sigma}{\sqrt{n}}\right)$$

Solved Example – Z-Test CI

Problem:
A sample of 100 students has an average score of 75. The population standard deviation is 10. Find the **95% confidence interval** for the true mean.

Solution:

$$\bar{x} = 75, \quad \sigma = 10, \quad n = 100, \quad Z_{0.025} = 1.96$$

$$CI = 75 \pm 1.96 \times \left(\frac{10}{\sqrt{100}} \right) = 75 \pm 1.96 \times 1 = 75 \pm 1.96$$

95% CI = (73.04, 76.96)

Solved Example – t-Test CI

Problem: A sample of 16 patients has an average blood pressure of 120. The sample standard deviation is 8. Find the **95% confidence interval** assuming a t-distribution.

$$CI = \bar{x} \pm t\left(\frac{s}{\sqrt{n}}\right)$$

Solution:

$$\bar{x} = 120, \quad s = 8, \quad n = 16, \quad df = 15, \quad t_{0.025,15} \approx 2.131$$

$$CI = 120 \pm 2.131 \times \left(\frac{8}{\sqrt{16}}\right) = 120 \pm 2.131 \times 2 = 120 \pm 4.262$$

95% CI = (115.74, 124.26)

p-values

What is a p-value in Hypothesis Testing?

The **p-value** (or probability value) tells us how likely we are to observe the data we have **if the null hypothesis (H_0) were true**.

In simple words:

It measures the **evidence against H_0**.

- **Small p-value** → strong evidence against H_0
- **Large p-value** → weak evidence against

Example:

A company claims their average delivery time is **30 minutes**. You collect data and find the average is **34 minutes**. After running a hypothesis test, your **p-value = 0.01**. What does it mean?

Interpretation:

Ho: average delivery time = 30 mins

If H_0 (delivery time is 30 mins) were true, there's only a **1% chance** of getting a result like 34 mins or more extreme.

Now, Since **p = 0.01 < 0.05**, we **reject H_0**.

We conclude that the actual delivery time is **significantly different** from 30 mins.

MCQ

Q1. $p=0.03 \rightarrow ?$

A) Accept H_0

B) Reject H_0

Q2. CI=(48,52), $\mu_0=53 \rightarrow ?$

A) Reject H_0 when your population or sample mean is outside the CI, reject the null hypothesi

B) Accept H_0

Q3) $p=0.07$, $\alpha=0.05 \rightarrow ?$

A) Reject H_0

B) Fail to reject H_0

MCQ

Q1. $p=0.03 \rightarrow ?$

A) Accept H_o

B) Reject H_o ✅

Q2. $CI=(48,52)$, $\mu_o=53 \rightarrow ?$

A) Reject H_o ✅

B) Accept H_o

Q3) $p=0.07$, $\alpha=0.05 \rightarrow ?$

A) Reject H_o

B) Fail to reject H_o ✅

Session Summary

In this session, we have discussed about:

- Understand A/B testing
- Confidence interval role in hypothesis testing
- Concept of p-values in statistical tests

Reference Links

- How to perform A/B Testing in Python
- Introduction to Hypothesis Testing
- Frequently asked hypothesis testing interview questions and answers – part 1
- Frequently asked hypothesis testing interview questions and answers – part 2
- Frequently asked hypothesis testing interview questions and answers – part 3
- Sampling methods guide
- How to Collect Data Effectively: Methods, Questionnaire Design, and Data Sources Explained
- Probability Distributions in Python
- Step-by-Step Guide to Two-Sample Z-Test for Large Samples (Population SD Known)